First we create a **Student Class.**
All student properties are private (This is called **Encapsulation**)

We need a unique ID for every student.
Use – static int id = 1; (ID work as a ***counter***)

Then create a Student Constructor - A constructor is automatically called when we create an object . The constructor receives: name , age ,grade .

[Understanding this.studentId = id++]
This line automatically creates a unique - ID. this.studentId = id++;

Then Use Getters - Because the properties are `private`, we cannot access them directly.So we use **getter methods** to read the values.

Use Setters
We use setter methods to change private values.

Then Create GraduateStudent
(class GraduateStudent **extends** Student)
`extends` means **Inheritance**.

Add Research Topic

GraduateStudent needs one extra property:

private String researchTopic;
This stores the student's research topic.

Then again Create GraduateStudent **Constructor**
The constructor receives:name ,age, grade, researchTopic

We use:
super(name, age, grade);
`super()` calls the **Student constructor** and initializes the name, age, and grade.

Then:

this.researchTopic = researchTopic;

stores the research topic.

Finally, in the `main()` method, we create two Student objects and one GraduateStudent object. We display their information using getters, then update the student information using setters and display the updated information.